

- 1) Alter the table STUDENT2 with following structure.

```
mysql> ALTER TABLE STUDENT2
-> ADD PRIMARY KEY(RegNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> DESC STUDENT2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES	UNI	NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

6 rows in set (0.00 sec)

- 2) Alter the table name FACULTY with following structure.

```
mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY(FacNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> ALTER TABLE FACULTY
-> ADD CONSTRAINT chk_gender
-> CHECK (Gender IN ('M','F'));
ERROR 3822 (HY000): Duplicate check constraint name 'chk_gender'.
mysql> DESC FACULTY;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	NO	PRI	NULL	
FacName	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	bigint	YES		NULL	
Dept	varchar(10)	YES		NULL	

7 rows in set (0.00 sec)

- 3) Alter the table name DEPARTMENT with following structure.

```
mysql> ALTER TABLE DEPARTMENT
      -> ADD PRIMARY KEY(DepNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> DESC DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DeptNo	varchar(4)	NO	PRI	NULL	
DeptName	varchar(15)	YES		NULL	
DeptHead	varchar(4)	YES		NULL	

3 rows in set (0.00 sec)

- 4) Alter the table name COURSE with following structure.

```
mysql> ALTER TABLE COURSE
      -> ADD PRIMARY KEY(CourseNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> ALTER TABLE COURSE
      -> ADD CONSTRAINT chk_sem
      -> CHECK (SemNo IN ('1','2','3','4','5','6'));
ERROR 3822 (HY000): Duplicate check constraint name 'chk_sem'.
mysql> DESC COURSE;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(3)	NO	PRI	NULL	
CourseDesc	varchar(14)	YES		NULL	
CourseType	char(1)	YES		NULL	
SemNo	char(1)	YES		NULL	
HallNo	varchar(4)	YES		NULL	
FacNo	varchar(4)	YES		NULL	

6 rows in set (0.00 sec)

- 5) Alter the table name STUDENT2 with following structure.

```
mysql> ALTER TABLE STUDENT2
-> ADD CONSTRAINT uk_student
-> UNIQUE(S1Name);
Query OK, 0 rows affected, 1 warning (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 1

mysql> DESC STUDENT2;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
S1Name	varchar(15)	YES	UNI	NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

```
6 rows in set (0.00 sec)
```

- 6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

```
mysql> CREATE TABLE student3 (
->     RegNo INT DEFAULT NULL,
->     Marks INT DEFAULT NULL,
->     CONSTRAINT fk_regno
->     FOREIGN KEY (RegNo)
->     REFERENCES student2 (RegNo)
-> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
ERROR 1050 (42S01): Table 'student3' already exists
mysql> DESC STUDENT3;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	YES	MUL	NULL	
Marks	int	YES		NULL	

```
2 rows in set (0.00 sec)
```